FinTrack Pro - Simple Input & Output Specifications

Project Title:	Data-Optimized Student Expense & Budget Tracker	**Candidate Roll No:**	25A31A05ES
Backend Framework:	Java / Spring Boot REST API	**Data Layer Architecture:**	MySQL + Isolated In-Memory ADS Manager

This technical blueprint documents the exact structures, parameter requirements, and JSON payloads for the primary operational entry points within the FinTrack Pro architecture. These structures directly feed the custom Trie, AVL Tree, and Segment Tree engines.

1. User Authentication & Session Security Module

HTTP POST REQUEST MATRIX

User Registration Pipeline (/api/auth/signup)

Transfers raw client authentication criteria to the server engine. Plaintext values are instantly passed to BCrypt hashes.

```
{
    "fullName": "Talluri Jahnavi",
    "email": "jahnavi.25a31a05es@college.edu",
    "username": "jahnavi_es",
    "password": "SecurePassword123!",
    "confirmPassword": "SecurePassword123!"
}
```

HTTP JSON RESPONSE MATRIX

```
{
    "success": true,
    "message": "User registered successfully with default role: STUDENT",
    "assignedRole": "STUDENT",
    "timestamp": 1794171600000
}
```

HTTP POST REQUEST MATRIX

User Login Pipeline (/api/auth/login)

Establishes secure, stateful server-side sessions bound tightly to the user context. Ignores frontend-supplied user identifiers.

```
{
    "usernameOrEmail": "jahnavi_es",
    "password": "SecurePassword123!"
}
```

HTTP JSON RESPONSE MATRIX

```
{
    "authenticated": true,
    "username": "jahnavi_es",
    "role": "STUDENT",
    "sessionToken": "JSESSIONID_A83F92C1D74E5B",
    "redirectUrl": "/dashboard"
}
```

2. Expense Core Engine Module (AVL Tree & Trie Interaction)

HTTP POST REQUEST MATRIX

Add Transaction API Endpoint (/api/expenses/add)

Triggers database entry generation and simultaneously balance-adjusts the user's localized transaction AVL Tree and updates the Category Trie.

```
{
    "amount": 750.00,
    "category": "Hostel Mess Fee",
    "dateString": "2026-11-08",
    "description": "Monthly mess charge submission"
}
```

HTTP JSON RESPONSE MATRIX

```
{
    "transactionId": 1042,
    "amount": 750.00,
    "category": "hostel mess fee",
    "epochTimestamp": 1794162600000,
    "description": "Monthly mess charge submission",
    "status": "PROCESSED_AND_INDEXED_IN_AVL_TREE",
    "executionComplexity": {
        "databaseSave": "O(1)",
        "inMemoryTreeInsertion": "O(log N)"
    }
}
```

HTTP GET PARAMETERS MATRIX

Prefix-Based Category Autocomplete Lookup (/api/expenses/autocomplete)

Queries the isolated `ExpenseTrie` root instance associated with the active session.

- **Query Key:** `prefix` = "ho"
- **Security Filter:** Authenticated principal session token evaluated on backend.

HTTP JSON RESPONSE MATRIX

```
[
    "hostel rent",
    "hostel mess fee",
    "hospital charges"
]
```

HTTP GET PARAMETERS MATRIX

Temporal Range Filter Routing (/api/expenses/filter)

Executes bounded sub-tree queries utilizing fast in-order sweeps over the timestamp keys.

- **Param 1:** `startDate` = "2026-11-01" (Converted to epoch: 1793490600000)
- **Param 2:** `endDate` = "2026-11-08" (Converted to epoch: 1794181800000)

HTTP JSON RESPONSE MATRIX


```
[
    {
        "id": 1021,
        "amount": 150.00,
        "category": "canteen",
        "timestamp": 1793577000000,
        "description": "Afternoon snacks with friends"
    },
    {
        "id": 1042,
        "amount": 750.00,
        "category": "hostel mess fee",
        "timestamp": 1794162600000,
        "description": "Monthly mess charge submission"
    }
]
```

3. Budget Planning & Segment Tree Analytics Module

HTTP POST REQUEST MATRIX

Configure Monthly Cap Bounds (/api/budget/set)

```
{
    "month": 11,
    "year": 2026,
    "allocatedLimit": 5000.00
}
```

HTTP JSON RESPONSE MATRIX

```
{
    "budgetId": 84,
    "allocatedLimit": 5000.00,
    "totalSpentMonthToDate": 900.00,
    "remainingBalance": 4100.00,
    "exhaustionPercentage": 18.00,
    "segmentTreeStatus": "SYNCHRONIZED"
}
```

HTTP GET PARAMETERS MATRIX

Segment Tree Day-Range Aggregate Analysis (/api/analytics/range-sum)

Queries the balanced tree structure to isolate target day segments without linear record traversal.

- **Param 1:** `startDay` = 1
- **Param 2:** endDay = 7

HTTP JSON RESPONSE MATRIX

```
{
    "requestedRange": "Day 1 to Day 7",
    "aggregateSum": 150.00,
    "segmentTreeQueryComplexity": "O(log N)",
    "alertTriggered": false
}
```